

Rajat Aggarwal

📞 9646437049 | ✉️ rajataggarwal9646@gmail.com | 💻 [Rajat Aggarwal](#) | 🔄 [rajat552](#) | 📍 IND. Citizen

EXPERIENCE

MalariaGEN

2026 – Present

Open Source Contributor

Remote

- Constructed the Youtube_schema helper for the MalariaGEN **Python API** (PR #1124) to streamline sample **metadata introspection**.
- Utilized **Python** and **Pydantic** to validate complex **data models**, supporting large-scale open-source data workflows.

Boa JavaScript Engine

2025 – Present

Open Source Contributor

Remote

- Contributed to an experimental **Rust-based** JS engine, streamlining developer onboarding by improving **CLI documentation** via peer-reviewed pull requests.

Hackathon Participant

2024 – Present

Various National-Level Hackathons

Hamirpur, HP

- Competed in **Amazon Nova AI**, Electrothon 7.0, and Hack On Hills 6.0; engineered and presented rapid prototypes for **web-based AI** and **automation tools**.

PROJECTS

Gemini Copilot (AI Productivity Tool) | [Live Demo](#)

- Architected an **autonomous AI agent** to reason, plan, and execute workflows, aligning with **agentic enterprise** principles.
- Engineered a resilient **hybrid LLM backend**, integrating **Amazon Bedrock** (Nova 2 Lite) with an automated fallback to **Gemini 1.5 Flash**.
- Constructed robust **RESTful API** integrations using **Node.js** to parse complex data inputs and power real-time, AI-driven task automation.

Amazon-Clone-UI | [Live Demo](#)

- Developed a responsive **HTML5/CSS3** e-commerce clone, utilizing **client-side rendering** for complex hierarchical layouts.
- Optimized **DOM manipulation** and asset loading to achieve high-performance rendering and seamless cross-device navigation.

EDUCATION

National Institute of Technology (NIT) Hamirpur

Aug. 2024 – May 2028

Bachelor of Technology in Computer Science Engineering

Hamirpur, Himachal Pradesh

- Coursework:** Data Structures, Algorithms, Programming in C/C++, **Python**, Computer Fundamentals, Discrete Mathematics
- Activities:** Executive Member at National Service Scheme (NSS), Bhangra Club

SKILLS

Languages: Python, SQL, C++ (DSA), TypeScript, JavaScript (ES6+)

Frameworks/Web: Node.js, Express.js, Next.js, React.js

Backend/Data: RESTful APIs, pandas, LLMs (Gemini), JSON Schema, MongoDB

Tools/DevOps: Git, GitHub (CI/CD workflows), Linux (Kali/Ubuntu), Technical Documentation